OBJECT ORIENTED SOFTWARE ENGINEERING

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Understanding Requirements

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- The broad spectrum of tasks and techniques that lead to an **understanding of requirements** is called **requirements engineering.**
- Requirements engineering builds a **bridge to design and construction.**
- The most difficult tasks that a software engineer used to face is **understanding the requirements of a problem.**

- Requirements engineering tasks lead to an understanding of :
  - What the **business impact of software** will be
  - What the **customer wants** and
  - How **end users will interact** with the software

- Software engineers or "**analysts**" and other **project stakeholders** (like managers, customers, end users) **all participate** in requirements engineering

There are Seven **distinct task** in Requirements Engineering.
They are:

1. **Inception,**
2. **Elicitation,**
3. **Elaboration,**
4. **Negotiation,**
5. **Specification,**
6. **Validation, and**
7. **Management.**

1. **Inception**—It is a task that defines the **scope & nature** of the problem to be **solved.**

- Inception is a task where the requirements engineering **asks a set of questions** to establish a software process.
- In this task, it understands the problem and evaluates with the proper solution.
- It collaborates with the relationship between the customer and the developer.
- The developer and customer decide the **overall scope** and the **nature** of the question

For example:

✓ask a set of questions that establish …

- Basic understanding of the problem.
- The people who want a solution.
- The nature of the solution that is desired, and
- The effectiveness of preliminary communication and collaboration between the customer and the developer.

2. Elicitation—
A task that helps stakeholders **define what is required**.

- Elicitation means to find the requirements from user.

- The requirements are difficult because the **following problems occur in elicitation**.

(i). Problem of scope: The customer give the unnecessary technical detail rather than clarity of the overall system objective.

(ii) Problem of understanding:

Poor understanding between the customer and the developer regarding various aspect of the project like **capability, limitation** of the computing environment.

(iii). Problem of volatility:

In this problem, the requirements change from time to time and it is difficult while developing the project.

3. Elaboration—create an analysis model that identifies data, function and behavioral requirements.

- In this task, the information taken from user during inception and elicitation and are expanded and refined in elaboration.
- Its main task is **developing pure model of software** using functions, feature and constraints of a software.

4. Negotiation—agree on a deliverable system that is realistic for developers and customers

- In negotiation task, a software engineer decides that how will the project be achieved with limited business resources.
- To create rough guesses of development and access the impact of the requirements on the project cost and delivery time.

Eg:

  - what are the **priorities ?**
  - what is essential ?
  - when is it required?

5. Specification

In this task, the requirements engineer constructs a final work product.

- The work product is in the form of software requirements specification.

- In this task, formalize the requirements of the proposed software such as informative, functional and behavioral.

- The requirements are formalize in both graphical and textual formats.

- **In specification,** finally, the problem is specified in some manner and then **reviewed ,** can be any one (or more) of the following:
  - A written document
  - A set of models
  - A formal mathematical notations
  - A collection of user scenarios (use-cases)
  - A prototype

6. Validation -

- A Validation is a **review mechanism** that looks for:
  - errors in content or interpretation
  - areas where clarification may be required
  - missing information
  - inconsistencies (a major problem when large products or systems are engineered)
  - conflicting or unrealistic (unachievable) requirements.

6. Validation…

- The work product is built as an output of the requirements engineering and that is **accessed for the quality** through a validation step.

- The formal technical reviews from the software engineer, customer and other stakeholders helps for **the primary requirements validation mechanism**

7. Requirements management

- It is a set of activities that help the project team to **identify, control and track** the requirements and changes can be made to the requirements at any time of the ongoing project.

- These tasks start with the identification and assign a unique identifier to each of the requirement.
- After finalizing the requirements, traceability table is developed.

- The examples of traceability table are the **features, sources, dependencies, subsystems and interface** of the requirements

THANK YOU

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